

MATTIA DANESE

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WORK EXPERIENCE

Full Stack Software Engineer @ Hexmodal — *Python (Django), Vue.js, AWS* Sep 2025 – Present

- Designed and built a resilient IoT device test orchestration system for the MicroInverter product, implementing multi-phase Celery task workflows with automatic retry logic and real-time UI progress tracking — eliminating the need for on-site manual device intervention
- Consolidated the ELight and MicroInverter products into a unified device type, refactoring backend models, API serializers, URL routing, and frontend views to simplify device taxonomy across 120,000+ devices and 2,000+ clients in 48 states
- Implemented the ELight anti-tampering detection system end-to-end, from LoRaWAN uplink parsing and device state management through visual inspection workflows and the detail-page status UI
- Shipped features, hotfixes, and infrastructure improvements across the full stack and all 6 IoT product lines, constantly context-switching between products and working directly with the Product and Ops teams in a fast-paced Series A environment

Software Engineer @ Emerest — *Python, TypeScript, C#/.NET, AWS* Jan 2024 – May 2025

Dual role across Data Engineering (sole developer) and Backend Platform teams

- Architected and managed a MongoDB database storing critical patient data, along with the corresponding ingest pipelines powering client-facing reports that tracked patient-caregiver interactions
- Built 20+ automated data pipelines powering daily and weekly executive reports used by upper management to analyze platform feature adoption and user engagement
- Developed 16 client-facing API endpoints powering in-app caregiver reports on patient engagement in weekly platform meetings, supporting JSON and CSV responses with client-side customization
- Refactored 30+ existing AWS Lambda functions into a hierarchical class architecture with private modules, improving maintainability and code reuse across the platform

Full Stack Software Engineer Intern @ MassEnergize — *Python (Django), React.js* 2022

Full Stack Software Engineer Intern @ WAISN — *Python (Django), React.js* 2021

PROJECTS

echo — text-echo.com — *Next.js, React.js, TypeScript, PostgreSQL, Drizzle ORM* Winter 2025

- Text-based social media platform enabling users to share and discover songs through friend networks
- Integrated Twilio API for text-based UX (song sharing, curated playlists) and Trigger.dev for background async jobs (playlist generation, incoming message handling)
- Implemented cross-platform song matching between Spotify and Apple Music using native APIs and SongLink API to resolve track identities across streaming services

Listen Up — *Python (Flask), Twilio API, Google News API, ChatGPT API* 2023

- Voice-based news app that converts spoken queries into real-time article delivery via phone calls, using voice-to-text transcription, Google News API search, and on-demand web scraping with local caching to avoid redundant fetches
- Reduced article processing time from 6.5s to 0.5s by parallelizing scraping and summarization with multithreading
- Led a team of 3 in architecting the Flask backend to orchestrate the full pipeline: transcription, search, scrape, summarize, and text-to-speech playback

Image Compressor & Decompressor — *C* 2020

- Command-line tool for lossy image compression using low-level bit-packing modules and RGB compression algorithms

TECHNICAL SKILLS

Languages: Python · C/C++ · C#/.NET · Java · JavaScript/TypeScript · Go

Frameworks: Django · React.js · Next.js · Vue.js · Node.js · Flask

Tools & Infrastructure: AWS · Docker · Git · PostgreSQL · MongoDB · MySQL · Redis · Celery · Sentry

PERSONAL

US-Italy Dual Citizen · Italian (fluent) · Spanish (conversational) · Inter Milan · Skiing · Chess · Personal Finance · Traveled to 14 Countries

EDUCATION

Tufts University, Gordon Institute — *MS Computer Science* 2023

Tufts University, School of Engineering — *BS Computer Science* 2022